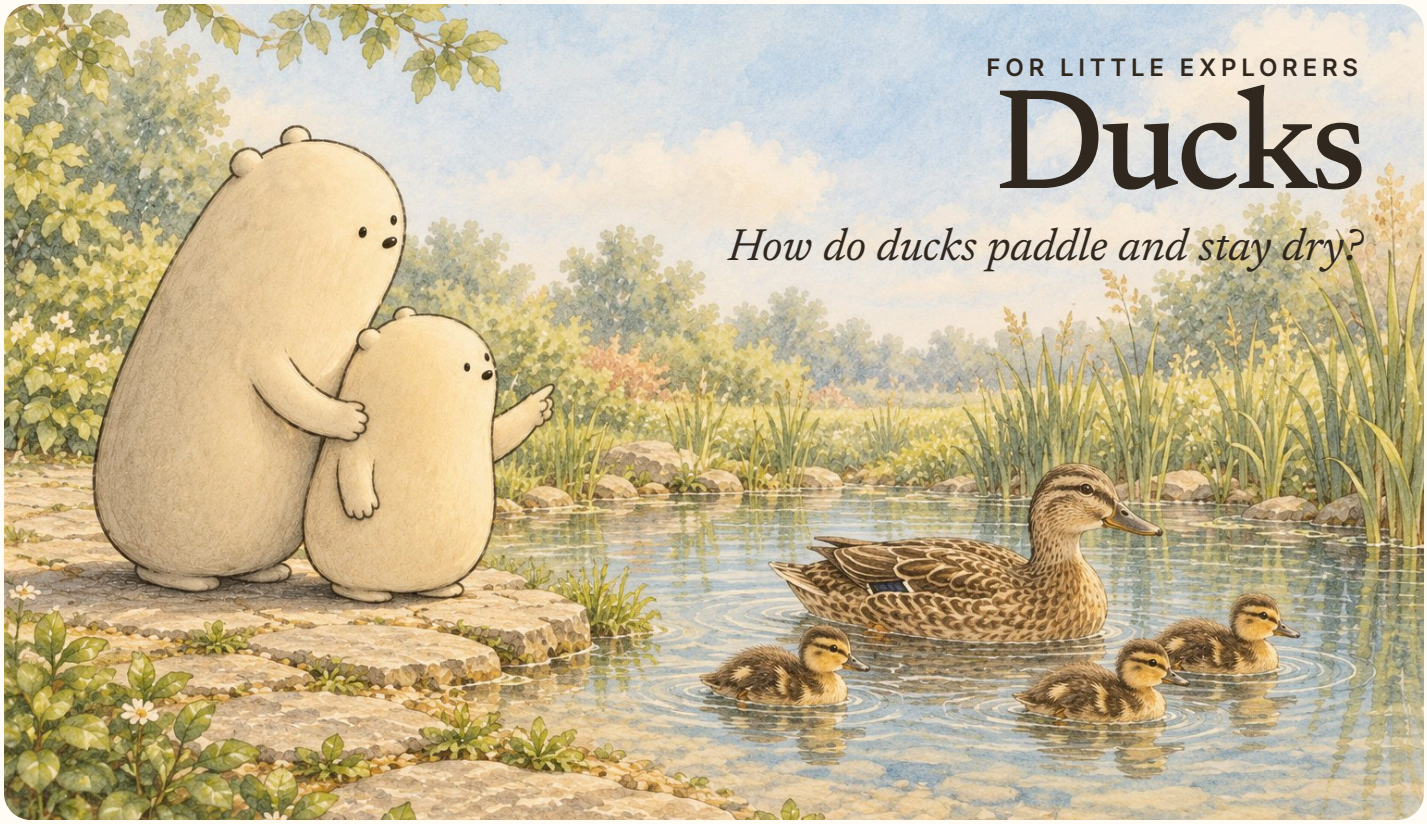


FOR LITTLE EXPLORERS

Ducks

How do ducks paddle and stay dry?

READ ALOUD

Quack. Paddle. Swish. (Move your hands like gentle paddles.) *Big Wonderer and Little Wonderer stand together on a dry path beside a shallow pond. A duck family glides past.*

A duck's feathers overlap like tiny roof tiles. They make a smooth coat around the duck's body.

ASK **Can you point to the feathers along the duck's back?**

The duck reaches its bill toward a special oil near its tail. Then it smooths the oil over its feathers.

Brush, brush, brush.

Water beads up and rolls away. Air held among the feathers helps the duck stay warm and buoyant while the water supports its body.

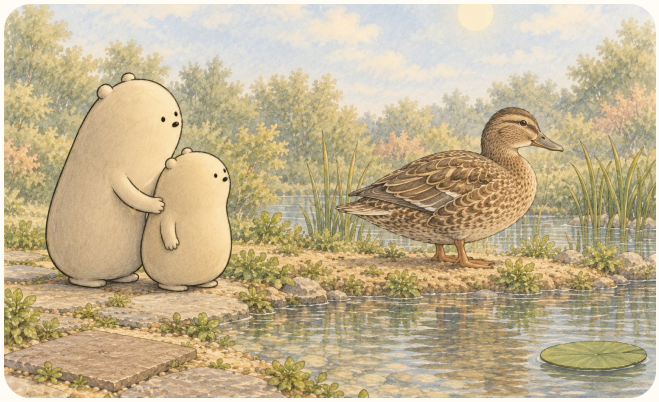
Under the water, wide webbed feet press backward. The duck's body moves forward.

ASK **Can your hands open wide like webbed feet?**

The duck family tucks close near the reeds. Big Wonderer and Little Wonderer watch quietly from the path.

Inside: a duck story · feather and foot science · first-then-last · spot the difference · a maze · counting to 20 · shapes and colors · rhymes and letter sounds · tracing and snipping · a coloring page · and a fold-up picture book.

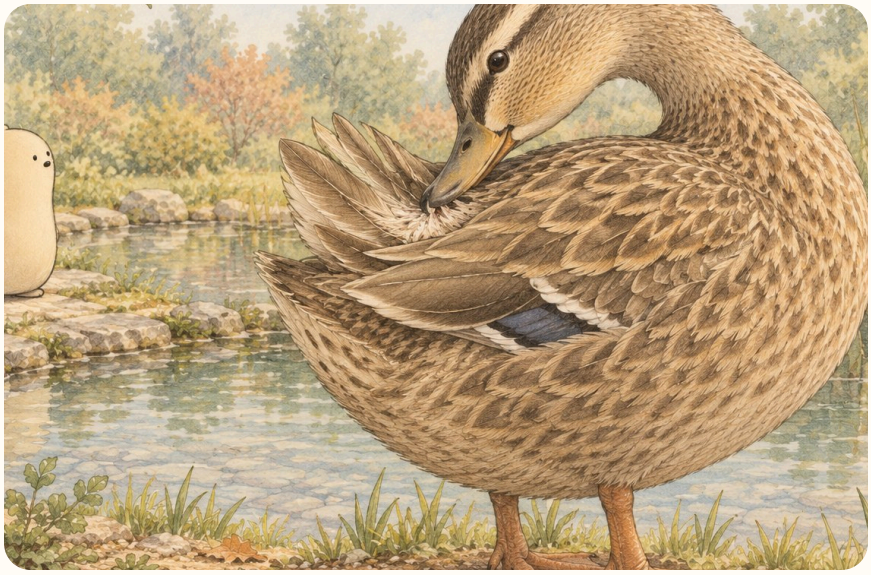
A duck gets ready to paddle



Feathers overlap

A duck's feathers overlap like tiny roof tiles. Together they make a smooth coat around the duck's body.

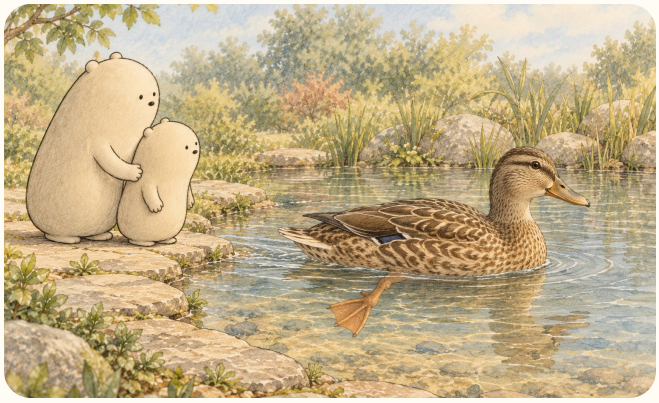
ASK Trace the feather rows with your finger.



The duck preens

A duck uses its bill to collect a little oil near its tail. It smooths the oil across its feathers. This careful feather care is called preening.

ASK Can you find the duck's bill touching its feathers?



Webbed feet push

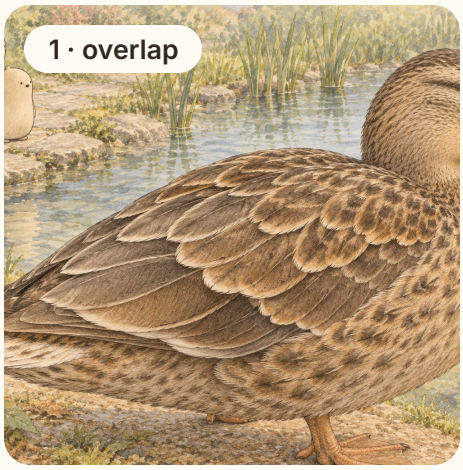
In the shallow pond, a duck spreads its webbed feet and pushes water backward. That push moves the duck forward.

ASK Show which way the duck is moving.

TRY IT WITH A GROWN-UP First, stand on a dry path with your grown-up. Then, watch a duck from far away without feeding or chasing it. Last, notice when its feet paddle.

A duck's water-ready body

Feathers, air and feet work together. They help a duck shed water, float and move across a pond.



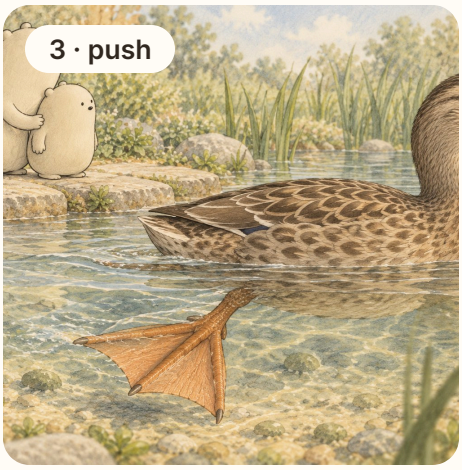
Feathers fit together

Rows of feathers overlap. Their tiny branches can hook together, making a smooth outer layer.



Drops roll away

Preening keeps the feathers neat and spreads oil. Water gathers into drops that can roll off.



A wide foot paddles

A duck opens its webbed toes for a wide push, then folds them closer as the foot swings forward.



Air helps, too

A duck's body and its fluffy feather layers hold air. With water pushing up, that trapped air helps the duck float.

ASK **Point to the duck riding high on the water.**

(Ducks still get wet around their feet and bills. Their feather coat protects the body underneath.)

Made for the water. *A smooth feather coat sheds water. A buoyant body rides near the surface. Webbed feet push against the water. Each part has a job.*

EXPERIMENT WITH A GROWN-UP First, lay three paper ovals so they overlap like feathers. Then, slide a finger from the top oval toward the bottom. Last, notice how one smooth layer covers the gaps.

Which duckling goes with which duck?

Three ducks and three ducklings have matching feather colors.

1 Use your finger to join each duckling to the duck with the same color.



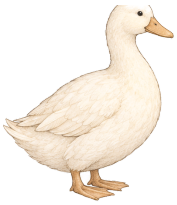
a white duckling



a gray duck



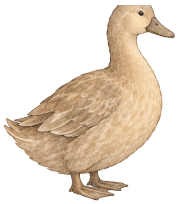
a brown duckling



a white duck



a gray duckling



a brown duck

WHY? These picture pairs match by color. Real duck families can have many different feather colors and markings.

2 Shapes and colors by the pond

Point to each shape, then say its color with your grown-up.



yellow circle

The round sun looks like a circle.



blue triangle

A folded wing can make a triangle shape.



brown square

A path stone can be a square.



green oval

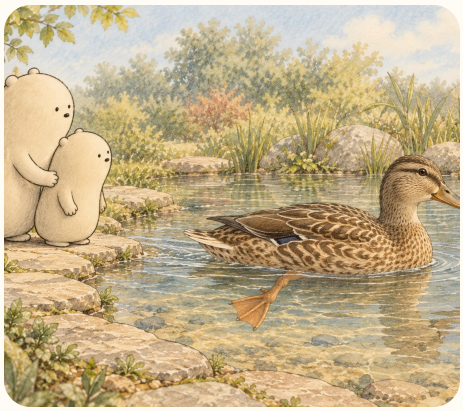
A lily pad can look like an oval.

ASK Can you find something round in your room?

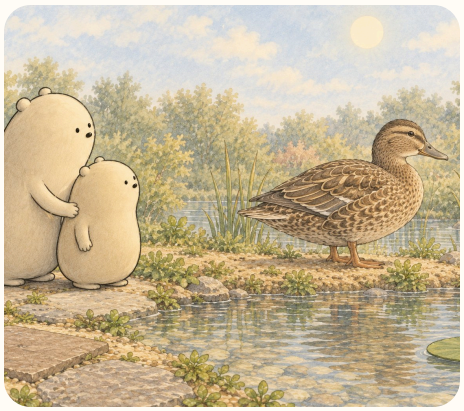
What happened first?

Grown-up: point to each picture and tell it in a few words. Tell the story together using First, Then, and Last. Little explorer circles 1, 2 or 3 under each picture.

1 Ready for the pond



1 2 3



1 2 3



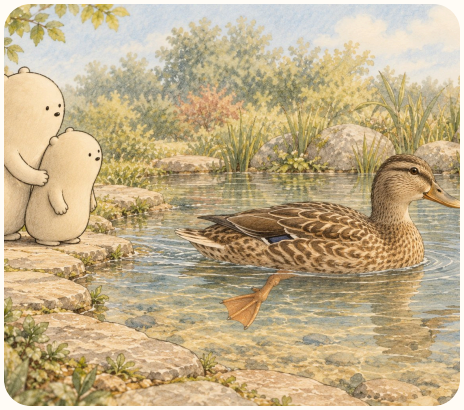
1 2 3

First, feathers overlap. Then, the duck preens. Last, webbed feet paddle.

2 Across the shallow pond



1 2 3



1 2 3



1 2 3

First, the duck paddles. Then, the family glides. Last, they rest near reeds.

TALK ABOUT IT What might a duck do after it climbs onto the shore?

Spot 5 differences

The two pictures look the same, but 5 things changed. Point to each one, then circle it in the bottom picture. Color a dot for each one you find.

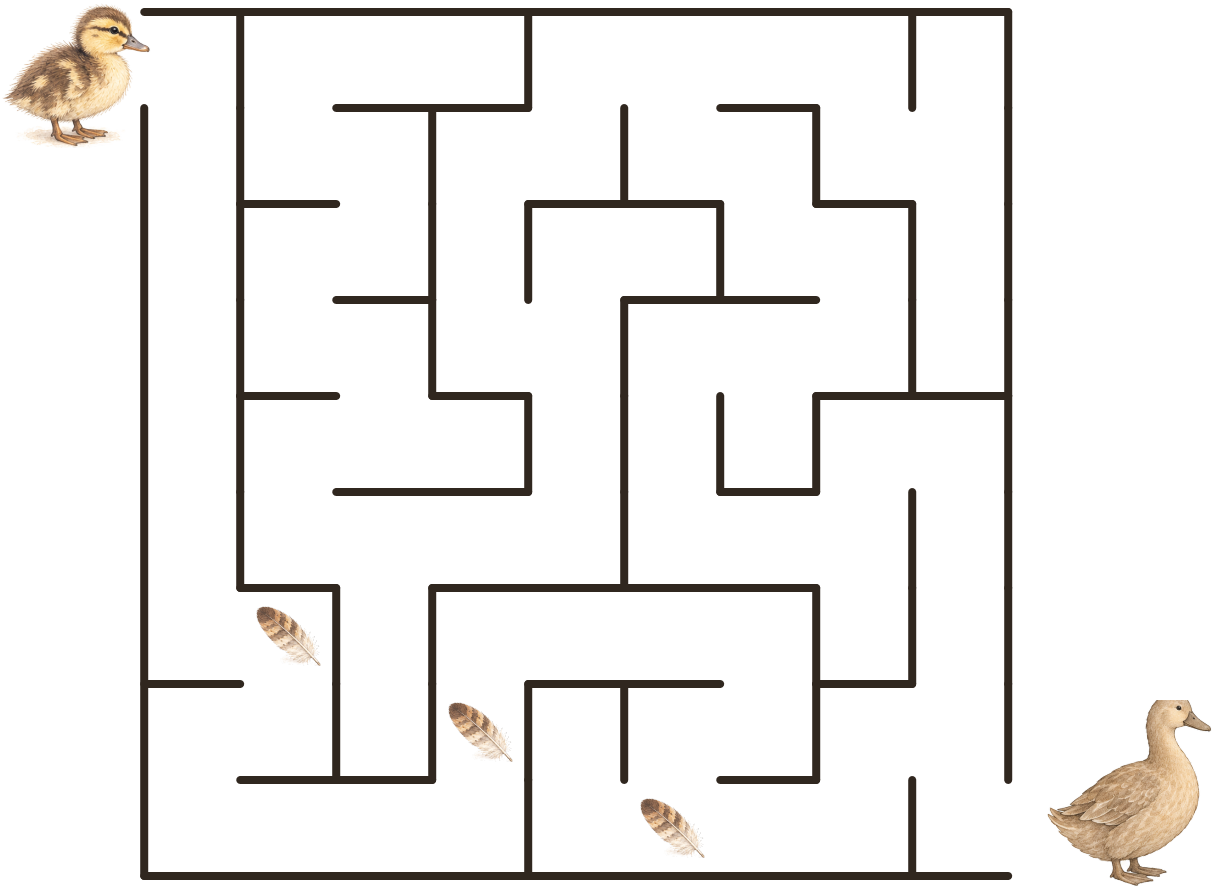


I found: ○ ○ ○ ○ ○

Follow the feather path

A duckling is following the dry path to the brown duck. Help it find the way and spot the three feather pictures.

First, trace the way with a finger.
Then, touch each feather as you pass it.
Last, follow the way again with your finger.



Feathers spotted:

WHY ARE DUCK FEET WEBBED? Skin joins a duck's front toes. The wide foot pushes a larger scoop of water backward, helping the duck move forward.

Duckling count

1 Count the ducklings. Circle how many.



Touch each duckling once as you count. Put a crayon dot on each duckling you touch.

7 8 9 11

2 Count the feathers. Circle the number.

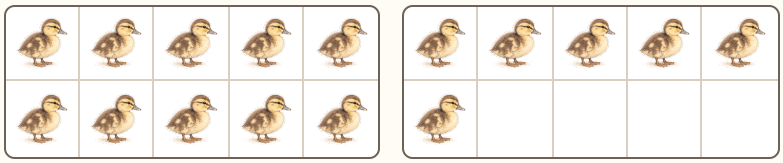
One full frame is 10. Then count on: 11, 12…



11 13 15

3 Count the ducklings. Circle the number.

10 and some more.



14 16 18

4 Find the number 14. Circle it.

8 12 14 17 20

5 Point to the group with MORE.



Pond picture stories

A stretch, just for fun. Grown-up reads each little story. Count the first group, then count on with the second group. Little explorer circles the answer.

3 ducks glide near the reeds. 2 more ducks join them. How many ducks now?



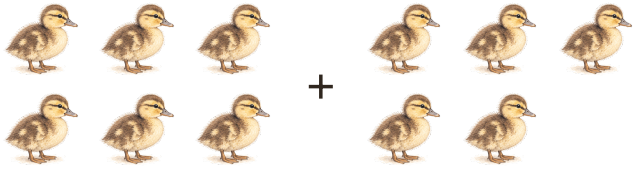
3 + 2 = (4) (5) (6)

4 feather pictures rest on one card. 3 feather pictures rest on another. How many feathers?



4 + 3 = (6) (7) (8)

6 ducklings paddle by a lily pad. 5 ducklings paddle near the reeds. How many ducklings?



6 + 5 = (10) (11) (13)

10 feather pictures fill the frame. 4 more feather pictures join them. How many feathers?



10 + 4 = (13) (14) (16)

Sounds by the pond

D d

This is the letter D, d. It says /d/, like the start of duck. Point to each D, d together.

1 **Which pictures start with /d/, like *duck*? Circle them.**


duck

sun


dog


cat


doll


bee

2 **Rhyme time! Grown-up says the three words. Point to the two that rhyme.**


duck


truck

sun


dog


log


bee


cat


hat


duck

3 **Read it together. Point to each word as you say it.**



A dog sits on a log.





The cat sits in the sun.

4 **Just for fun: touch each letter, then say the word fast.**



d o g



c a t

Trace and snip

Count the pictures together. Trace the first number with a finger, then trace the dotted numbers with a crayon. Start at the dot.



1 1 1 1



2 2 2 2



3 3 3 3



4 4 4 4



5 5 5 5

Snip along the pond breeze

With a grown-up helping, snip along the wavy line from the feather to the duck. Safety scissors, and a grown-up holds the paper.





Color the duck pond

First, color the round sun yellow.
Then, color the big oval lily pad green.
Last, circle the square path stone.

Try to stay inside the lines. Big and Little Wonderer can be any color you like.

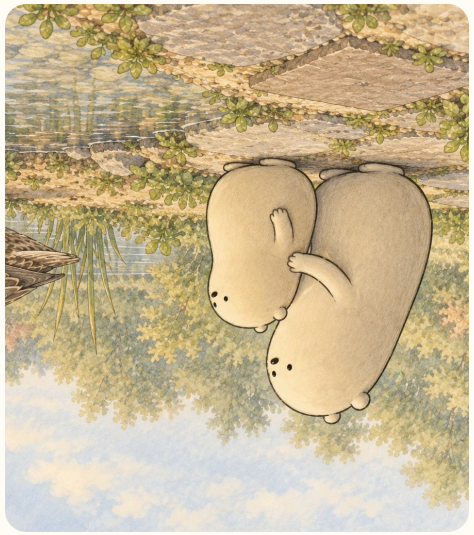


Ducks

FOR LITTLE EXPLORERS



How do ducks paddle and stay dry?



Feathers overlap like tiny roof tiles.



A duck preens and smooths its feathers.



Tiny feather branches fit together.



Water beads up and rolls away.



The duck family rests near the reeds.



That push moves the duck forward.



A wide webbed foot pushes water back.

How to fold the picture book

1. Print the picture sheet on one page, landscape, at 100% (no scaling).
2. Fold it in half the long way, then open it. Fold it in half the short way, then in half again; open it flat.
3. **Grown-up only:** fold it in half the short way and cut along the middle fold from the folded edge to the center crease.
4. Open it, fold it the long way again, and push the ends together so the slit opens into a little book. Page 1 goes on the front.

Read it together: one picture, a few words on each page. Ask "what do you see?" before you turn the page.